

Plan Conversion QA & Automation Program – WhiteCap Ascensus

1. Overview / Background

WhiteCap (Empower white-label) is onboarding multiple government-sponsored 529 plans to Ascensus. Their portal performs SSO handshake redirects into our SNSS portals (Member, CSR). After redirect, everything runs on our platform — which means **Plan Metadata, Funds, Tranche Setup, API Feeds, Kofax, Back-office Jobs, CSR, Member Portal flows all must be validated end-to-end.**

This project is NOT UI-heavy — the real risk lies in:

- Incorrect plan metadata
- Wrong traunch/fund mapping
- Broken Nightly/Recon/Price/Performance/Aggregation feeds
- Kofax ingestion errors
- API correctness for new PlanIDs
- Backoffice daemons & scheduled jobs
- CSR flows across 40+ features
- Member portal login through SSO handshake

We **cannot automate 100%** — but we have *major reusable components* already built.

The goal: **Deliver a reliable, targeted automation regression for this new Plan Conversion using existing frameworks + new tests where needed.**

2. Scope Summary (Automation vs Manual)

Below is the **clean, architect-approved mapping** of all modules you listed into:

- **Automation-ready** (existing framework or high ROI)
- **Manual mandatory** (edge cases, SSO handshake, low ROI)
- **Hybrid** (API automation + manual UI validation)

3. Automation vs Manual Matrix (Cleaned, Categorized)

A. Core Metadata & Backend

Story	Note	Automation/Manual (Need to evaluate)
Plan Metadata	API + DB	UI metadata screen sanity (Automated)
Funds Metadata	API + DB	Cross-plan manual(Automated)
Price & Performance	Backoffice job validation	CSR views
Contribution Metadata	API	UI sanity (Automated)
Withdrawal Metadata	API	UI sanity (Automated)
Flywire Metadata	API / File validation	Portal sanity
DBR Metadata	DB jobs	UI
Statements & IRR	API/file presence	UI formatting
Backoffice Jobs (ACH, Recon, Max Contrib, Account Sync)	DB + file checks	Functional E2E
Age-based Roll Daemon	DB checks	UI
IDP Metadata	DB validation	UI only

B. APIs – Best Automation ROI

Story	Automation	Manual
Account – POST / PUT / GET		–

Metadata – GET		–
Plan – GET		–
Allocation – GET		–
Kofax Feeds	batch validations (Partial Automation exists)	File handling
File Validations (Mellon/Recon/Price/Flywire)		file handling

C. UI – Member/CSR (most modules already automated)

Below is what **we already have** automated for existing plans — meaning **50–70% reusable**:

CSR Area	Automation	Manual
CSR Login		–
Account Overview		–
As of Balance		
Graph		
Download Transactions		–
Transaction History		–
Change Txn Year		–
CSR Transaction: Contribution		–
CSR Transaction: Withdrawal		–
CSR Transaction: Exchanges	Need to be added to Nightly	–
CSR Transaction: Transfer	Need to be added to Nightly	–
CSR Transaction: Fees/Reversals	Need to be added to Nightly	–
CSR Transaction: Ugift	Need to be added to Nightly	–
CSR – Share Adjustment	Need to be added to Nightly	–
CSR – Principal Update	Need to be added to Nightly	–
CSR – Information Pages	Need to be added to Nightly	–
CSR – Aggregation/OFAC		
CSR – Maintenance History		
CSR – Member Actions		
CSR – Make Contribution / Recurring		–
CSR – Rollover		
CSR – Change Investment	Need to be added to Nightly	–
CSR – Profile & Docs		–
CSR – Owner Profile		–
CSR – Beneficiaries		–
CSR – Authorized Agents		–
CSR – Trusted Contact		–
CSR – Image Upload		
CSR – Right-hand side functionality	Need to be added to Nightly	–

Not all but most of these scripts exist. They require:

- Data refresh
- New PlanID branching

- Traunch/Investment mapping updates
- Regression tuning for failed cases
- May some of them will need rework done (not all added to regression suite)

ROI = Extremely high.

D. Member Portal – Hardest part

Because of **WhiteCap SSO Ascensus** handshake.

Area	Automation	Manual
Enrollment	(post handshake)	handshake part
Save for later		–
Login	SSO handshake MFA Disabled accounts needed	Manual
Forgot Password / Username	IDP MFA Disabled accounts needed	Manual
MFA		Manual
Profile Updates		–
Contribution flows		–
Withdrawals		–
Transfer Beneficiary		–

Automation will begin **after SSO redirect**.

E. Kofax, Gifting, Payroll, Files

All these can be automated using DB + file polling.

4. Phase-Wise Implementation Strategy

PHASE 1 – High-ROI Automation (4–6 weeks)

Use what already exists.

Deliverables

1. CSR regression suite for new PlanID
2. API suite for POST/PUT/GET
3. Metadata validations (DB + API)
4. Core Backoffice tests:
 - Price
 - Performance
 - Recon
 - Max contrib
 - Account sync
5. File validations (Mellon/Recon/Price)
6. Kofax ingestion smoke

Effort

- 1 senior automation: ~120 hours
- 1 mid automation: ~140 hours
- 1 manual QA for metadata + SSO: ~40 hours

Note: added a 1 cushion for new resource (considering Onboarding & KT)

Total Phase 1 = 300 hours

PHASE 2 – Medium-Complexity (2-2.5 weeks)

1. Member portal automation (post-handshake only)

2. Advanced CSR flows (multi-txn, reversals, edge cases, Plan based expansion)
3. Payroll / Rollover forms
4. Gifting (Ugift hub)
5. File-based automation expansion
6. Plan Aggregation
7. Confirms
8. Mobile and Deep Linking
9. Equity Wash Restriction
10. Dormancy

Total Phase 2 = 160 hours

Ongoing PHASE – Manual Mandatory Areas (Ongoing)

1. SSO handshake WhiteCap Ascensus
2. Content checks (logo, footer, page text, links)
3. Cross-device testing (mobile, tablets)
4. Visual validations
5. IDP-based login scenarios

5. Staffing Recommendation

To finish before deadlines and without burdening Swapnil or Pramod:

Role	Count	Duration
Senior SDET	1	Full 3 months
Mid-level SDET + Manual QA	1	Full 3 months
DB/Backoffice/DevOps SME	0.5	On demand

Swapnil + Pramod = oversight?

6. What Automation Already Exists (Reusable)

Full CSR UI regression
 API framework
 Kofax partial framework
 X Backoffice file polling & validation scripts
 X DB validation utilities
 Member portal partial flows (post login/SSO)
 Enrollment automation